Testing Physics Regularization for Event-Camera Vibration Estimation Under Sensing-Condition Shift

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***Abstract*—Event cameras can observe structural vibration without contact at microsecond resolution, but it is unclear whether learned event-to-waveform estimators remain accurate when illumination or sensor settings change. We study this on 30 hammer-impact free-decay trials of an H-beam (XJTU-DV-Beam), grouped into six sensing conditions, with a laser Doppler vibrometer (LDV) as reference.**

We define a robustness gap Δ, the change in error between held-out and matched conditions, and a pre-specified falsification rule: a single-degree-of-freedom (SDOF) physics residual counts as helpful only if its Δ reduction exceeds trial-to-trial uncertainty. Using trial-level leave-one-condition-out evaluation, we compare a 3D-ResNet, a Kinetics-pretrained space-time Transformer, and a ridge regression baseline.

The physics term was not identifiable on four of six control folds; on the two folds with adequate precision the point estimate was unfavourable. It raised the matched-condition RMSE point estimate on most folds (not tested for significance). No model recovered waveform shape (Pearson $r \leq 0.25$ against a target above 0.9). Deep models reached matched RMSE of 0.19–3.08 across folds, against a window-level zero-prediction RMSE of 1.71–2.05 computed on the three high-variance folds only (On_set1, Off_set1, Off_set2; not computed for the other three), while ridge regression reached 0.29–0.66 by capturing amplitude envelope only.

Probing showed that latents encode envelope decay but not natural frequency. Shifted conditions were often easier than matched ones, indicating that the matched reference is unstable at these sample sizes.

We contribute the Δ protocol, an event-to-LDV pipeline, a normalised-residual self-consistency score, and a failure analysis that identifies the input representation, not the prior, as the bottleneck, with ridge regression as a robustness reference for future event-based vibration estimation.

***Index Terms*—event camera, non-contact vibration measurement, structural health monitoring (SHM), physics-informed learning, robustness gap**

I. Introduction

Vibration is the primary observable in structural health monitoring (SHM): changes in stiffness, mass or damping shift natural frequencies and reshape the dynamic response of bridges, beams, turbines and machinery. Each common sensing option has a drawback. Bonded accelerometers add mass and wiring and so alter the dynamics they measure. Laser Doppler vibrometers (LDVs) are accurate but need a stationary head and a clear line of sight. Frame cameras are contactless but blur on fast, small motion. A lower-cost, non-contact optical sensor that tolerates field lighting would ease inspection and continuous monitoring.

Event cameras are a candidate. Each pixel asynchronously reports a brightness change when $|\Delta \log I|$ exceeds a contrast threshold C, with microsecond timestamps and roughly 140 dB dynamic range, against about 60 dB for conventional CMOS sensors [1], [2]. These properties suit vibration measurement in principle.

The sensor itself is a source of domain shift. The effective contrast threshold varies across pixels [1], and illumination and bias settings change event rates, so the same physical motion can produce different event streams. For waveform regression there is no standard definition of robustness to this shift, since classification accuracy does not apply. A single-degree-of-freedom (SDOF) free-decay residual is a natural candidate remedy. It constrains the output to a physically plausible form whatever the sensing configuration. With only tens of trials, however, the residual must be identified from the same scarce data, and it can reward trivial solutions.

This paper tests that remedy. We learn a mapping f_θ from an event stream E_s, captured under sensing configuration s, to the waveform $\hat{q}(t)$ that a co-located LDV would record, and we ask three questions.

RQ1 (accuracy): Under matched acquisition conditions, can learned event-to-waveform estimators recover the LDV waveform and its dominant frequency?

RQ2 (stability): Does an SDOF physics residual reduce the robustness gap $\Delta = M(\text{shifted}) - M(\text{matched})$, with $M \in \{\text{RMSE, relative frequency error}\}$, by more than trial-to-trial uncertainty?

RQ2 is stated as a hypothesis with a pre-specified falsification rule: if the Δ reduction does not exceed its standard error, the physics term is reported as non-identifiable.

RQ3 (frequency vs waveform): Does the relative frequency error degrade less under sensing-condition shift than waveform RMSE?

Our contributions are:

- A continuous-waveform robustness protocol: the gap Δ with group-mean standard errors, trial-level leave-one-condition-out splits, and a falsification rule for regularisers.

- An event-to-LDV estimation pipeline (Struct-X) with a normalised-residual self-consistency score.
- An empirical test across a 3D-ResNet, a Kinetics-pretrained divided space-time Transformer and a ridge baseline: the physics term fails to clear noise on all six control folds and raises the matched RMSE point estimate on most folds, and no model recovers waveform shape.
- A failure analysis, supported by latent probing, that identifies the voxel representation as the bottleneck and ridge regression as a robustness reference for future work.

The study uses 30 hammer-impact free-decay trials of one H-beam rig (XJTU-DV-Beam [3]), grouped into six lighting/set conditions. The LDV reference is sampled at 256 Hz and the dominant mode lies near 6.7 Hz, so the sensor's kilohertz capability is not exercised here. We make no claim about damage classification or other structures.

The remainder of the paper is organised as follows: Section II reviews related work and the base paper, Section III the method, Section IV the experimental setup, Section V the results and discussion, Section VI the ablations, and Section VII concludes.

II. Related Work

Base paper. This study builds on the event-based visual deformation measurement of Wu et al. [4], which fuses events with co-registered frames and solves for dense displacement fields without training. It shows that events carry usable deformation information, but not that a learned event-to-waveform estimator transfers across sensing configurations.

That approach outputs deformation rather than an LDV-referenced waveform, requires a synchronised frame stream, is optimised separately for each sequence with no learned prior across recordings, depends on brightness constancy and spatial continuity, and was not evaluated under changes to the sensing configuration.

Two other literatures bear on the problem but, to our knowledge, have not been combined with event data. Physics-informed learning adds governing-equation residuals to data fitting [5], though a poorly specified residual can degrade accuracy [6]; in structural dynamics they improve data efficiency where mass, damping and stiffness are known or finite-element-derived [7]. The closest vision-based example fuses accelerometer and frame-camera displacement under a physics loss [8], but it inherits frame-camera blur on fast, small motion and applies the residual to the accelerometer channel rather than a high-bandwidth optical one. SHM research has also addressed environmental variation, but through classification under open-set domain adaptation [9], which does not carry over to a continuous waveform target. Its transferable lesson is that training and test data should differ by construction rather than by random split, which we adopt. Video transformers add a data constraint: they need pretraining to compete with convolutional models at small sample sizes [10].

III. Proposed Methodology

A. Problem Definition and Terminology

Let $E_s = \{(x_i, y_i, t_i, p_i)\}$ be an asynchronous event stream with timestamps t_i in microseconds and polarity $p_i \in \{-1, +1\}$, captured under sensing configuration s (illumination, contrast threshold C, bias currents, sensor pose). Let $q(t)$ be the displacement recorded by a co-located LDV. We learn

$$\hat{q} = f_\theta(E_s), \quad \hat{q} \in \mathbb{R}^T, \tag{1}$$

together with a dominant-frequency estimate $\hat{f}_n$. A condition is one lighting/set combination (On or Off $\times$ set 1–3), giving six conditions. Fold labels read as illumination (On/Off) plus acquisition set (set1–3, distinct sensor settings). In each fold, one condition is held out as the shift condition. The remaining five conditions are the matched conditions, used for training and for matched evaluation.

B. Models

Each encoder takes the voxel tensor as $B = 5$ bins with $C = 2$ polarity channels and feeds a waveform head ($T = 1000$ outputs) and a frequency head (softplus output). The three models we use are, from simplest to most expressive:

1) Ridge baseline.

We employ multi-output ridge regression with cross-validated regularization (RidgeCV [11]) on spatially averaged, per-channel temporal-bin means of the voxel grid ($d = 10$ features), fitted exclusively on training-trial windows. As a closed-form linear projection without learned representations, this model establishes a lightweight, deployable performance bound for evaluating deep architectures.

2) 3D-ResNet encoder.

To jointly capture local spatial and short-range temporal dynamics from voxel clips via inflated 3D kernels, we adopt an 18-layer 3D convolutional residual network (R3D-18 [12]). We adapt the ImageNet-pretrained input stem from three channels to two via weight averaging, mapping inputs to a 512-dimensional latent representation. End-to-end training is conducted without video-domain pretraining or self-attention mechanisms, serving as a lightweight convolutional benchmark.

3) TimeSformer encoder.

We utilize a 12-block divided space-time Transformer [10] that factorizes spatiotemporal attention by sequentially applying temporal self-attention across frames followed by spatial self-attention within frames. Initialized from Kinetics-400 weights [13] (evaluated strictly under CC BY-NC 4.0 research-use terms), the architecture is modified for two-channel input via stem weight adaptation, with temporal positional embeddings interpolated to B frames and patch tokens aggregated via mean pooling into a 768-dimensional latent vector.

The total loss is

$$L = L_{\mathrm{LDV}} + \lambda_p \cdot L_{\mathrm{phys}} + \lambda_s \cdot L_{\mathrm{smooth}} + \lambda_f \cdot L_{\mathrm{freq}} \tag{2}$$

with $\lambda_f = 0.1$ fixed. Its components are

$$L_{\mathrm{LDV}} = \mathrm{mean}_T \|\hat{q} - q\|^2,$$

$$L_{\text{smooth}} = \text{mean}_T \|d\hat{q}/dt\|^2,$$
$$L_{\text{freq}} = (\hat{f}_n - f_n)^2 / f_n^2.$$

The SDOF residual is

$$L_{\text{phys}} = \text{mean}_T \|\hat{q}'' + c\,\hat{q}' + k\,\hat{q}\|^2 \,/\, (\text{mean}_T \|\hat{q}\|^2 + \epsilon) \quad (3)$$

with $m = 1$ without loss of generality (the ODE can be divided by m); log-decrement fitting identifies only f_d and ζ, so absolute scale is set by L_{LDV}. The denominator was intended to guard against the collapse $\hat{q} \to 0$ trivially minimising the residual, but with the additive ϵ it provides no amplitude gradient for $\|\hat{q}\|^2 \gg \epsilon$ and rewards shrinkage near the ϵ floor, so it does not prevent collapse. It makes L_{phys} amplitude-blind by design, so amplitude is set by L_{LDV} alone. Derivatives use second-order central differences in the interior and first-order one-sided differences at the endpoints, computed in fp32 with $\epsilon = 10^{-6}$.

From one training trial per fold we estimate ζ by log decrement and the damped frequency f_d from the spectrum. Then $f_n = f_d/\sqrt{1-\zeta^2}$, $k = (2\pi f_n)^2$ and $c = 2\zeta\sqrt{k}$. Across folds $f_d \approx 6.7$ Hz, $k \approx 1800$ and $c \approx 2.2$–2.9. These values are frozen, and no validation, test or shift trial is used.

Four variants are trained: (a) data only, (b) smoothness only, (c) SDOF only, (d) both. Each physics-bearing variant selects (λ_p, λ_s) from the same 3×3 grid ($\lambda_p \in [10^{-3}, 10^{-1}]$, $\lambda_s \in [10^{-4}, 10^{-2}]$) by validation Δ_{RMSE}, with the same epoch budget. The frequency estimate $\hat{f}_n$ is the output of the trained head. The FFT peak of $\hat{q}$ is not used in reporting.

C. Training and Implementation

Models are implemented in PyTorch. We use Adam (learning rate 2×10^{-4}), micro-batch 8 with gradient accumulation to an effective batch of 16, at most 50 epochs, and early stopping with patience 10 on validation Δ_{RMSE}. A gradient-ratio controller tracks $r = \|\nabla L_{\text{phys}}\|/\|\nabla L_{\text{LDV}}\|$. It scales λ_p by 0.1 for the first five epochs, then halves λ_p if $r > 1$ for three consecutive epochs and doubles it if $r < 0.01$ for three consecutive epochs, within 10^{-4} to 0.1. Training is in full fp32.

IV. Experimental Setup

A. Data

We use the Beam subset of XJTU-DV [3], [14]: 30 hammer-impact free-decay trials of an H-beam, five per condition, with event streams in Prophese-style `.dat` files and LDV velocity in `.txt` files. Sensor resolution (1280×720) is read from the file headers. One event file carries a naming typo in the archive, which the pairing code handles. The LDV signal is sampled at 256 Hz, so it is integrated once to displacement and linearly detrended.

The pipeline runs in three sequential stages.

1) Event preprocessing.

For each trial we (i) mask hot pixels (event count above $5\times$ the median), (ii) drop same-pixel events less than 100 μs apart, (iii) crop to the contiguous y-band where the mean event count is at least 50% of its peak, plus 8 px padding, and (iv) build a voxel grid with $B = 5$ temporal bins and two polarity channels [1], giving a $(2, B, H, W)$ tensor (ten channels once bins and polarities are merged). The longest side is resized to 224 px and zero-padded to a square, preserving aspect ratio. Each channel is standardised with a mean and standard deviation frozen from training trials only ($\epsilon = 10^{-6}$).

2) Target preprocessing.

The displacement is low-pass filtered (4th-order Butterworth, cutoff $0.9\times$Nyquist $= 115$ Hz at the native 256 Hz rate), resampled to 1 kHz by interpolation (no new information), and z-scored with frozen training statistics, so the effective bandwidth is about 115 Hz. The dominant mode is near 6.7 Hz, giving 6–7 cycles per window. The frequency target f_n is the FFT peak of the full LDV trial (Hann window, $4\times$ zero-padding, parabolic interpolation; search 5–128 Hz).

3) Windowing.

Windows are 1.0 s long ($T = 1000$ samples) with a 0.5 s stride, anchored at the hammer trigger, taken as the peak of |velocity| (argmax, no hardware trigger). This is oracle alignment from the LDV target, not a deployable trigger, so reported accuracy is an upper bound. Windows starting less than 100 ms after the trigger are excluded, because the physics model assumes free decay. This gives 326 windows. A synthetic threshold sweep with v2e [15] was explored as augmentation-grade only and is not part of the reported results.

B. Data Characteristics

The 30 trials yield 326 windows, about eleven per trial. Every trial contains a single dominant mode near 6.7 Hz, so the frequency target has almost no spread and a constant predictor is hard to beat. Amplitude, by contrast, varies strongly across trials: one high-amplitude trial dominates the error on each of the three high-variance folds, which sets the effective sample size below the nominal 30 (quantified in Section V). The six conditions differ in event rate by construction (illumination on/off, three acquisition sets); that is the shift the protocol holds out, so no random split can stand in for it.

C. Split and Evaluation Protocol

Splits are made at trial level only; window-level splitting is not used. We use leave-one-condition-out evaluation with $K = 6$ folds. In each fold, the five trials of the held-out condition form the shift set, and the 25 trials of the other five conditions form the matched pool. The matched pool is shuffled and split 15/5/5 into train, validation-matched and test-matched. The shift trials are split two/three into validation-shift (tuning and early stopping only) and test-shift (final reporting only). The seed is fixed at 0, and a leakage check confirms that no trial appears in two sets. Each fold therefore yields one Δ, comparing five matched test trials with three shift test trials.

D. Evaluation Metrics

Let $q_w[n]$ and $\hat{q}_w[n]$, $n = 1 \ldots T$, be the reference and estimate on test window w ($T = 1000$, z-scored units), with

window means $\bar{q}_w$ and $\bar{\hat{q}}_w$. Per-window metrics are

$$\mathrm{RMSE}_w = \sqrt{(1/T)\sum_n (\hat{q}_w[n] - q_w[n])^2} \tag{4}$$

$$\mathrm{MAE}_w = (1/T)\sum_n |\hat{q}_w[n] - q_w[n]| \tag{5}$$

$$r_w = \frac{\sum_n (\hat{q}_w[n] - \bar{\hat{q}}_w)(q_w[n] - \bar{q}_w)}{\sqrt{\sum_n (\hat{q}_w[n] - \bar{\hat{q}}_w)^2 \cdot \sum_n (q_w[n] - \bar{q}_w)^2}} \tag{6}$$

$$\varepsilon_{f,w} = |\hat{f}_{n,w} - f_n| / f_n \tag{7}$$

We also record the normalised residual $\Phi_w = L_{\mathrm{phys}}$ (Eq. 3) evaluated on $\hat{q}_w$ as a self-consistency score. Pearson r is computed per window and never pooled across windows. By implementation, $r_w = 0$ when either signal is constant, so a collapsed model has $r = 0$ rather than an undefined value. For trial j with window set W_j, the trial-level value is

$$M_j = (1/|W_j|)\sum_{w \in W_j} M_w. \tag{8}$$

The trial is the unit of analysis. All means, standard deviations and standard errors are taken over trials, never over windows. For a group g of n_g test trials, $\bar{M}_g = (1/n_g)\sum_j M_j$ and $s_g^2 = (1/(n_g - 1))\sum_j (M_j - \bar{M}_g)^2$. Tuning and early stopping use Δ_{RMSE} over pooled validation windows, not the trial-level metric above.

Robustness gap. Trials are not paired across conditions, so Δ is a difference of group means:

$$\Delta_M = \bar{M}_{\mathrm{shift}} - \bar{M}_{\mathrm{matched}}, \quad M \in \{\mathrm{RMSE}, \mathrm{MAE}, \varepsilon_f\} \tag{9}$$

$$\Delta_r = \bar{r}_{\mathrm{matched}} - \bar{r}_{\mathrm{shift}} \tag{10}$$

with $\mathrm{se}_\Delta = \sqrt{s^2_{\mathrm{shift}}/n_{\mathrm{shift}} + s^2_{\mathrm{matched}}/n_{\mathrm{matched}}}$. The sign of Δ_r is reversed so that a positive Δ always means degradation. A negative Δ means the held-out condition scored better than the matched pool.

Pre-specified rule. For a given M, with "base" the data-only model and "phys" the physics-trained model,

$$R = \Delta_{\mathrm{base}} - \Delta_{\mathrm{phys}}, \quad \sigma_R = \sqrt{\mathrm{se}^2_{\mathrm{base}} + \mathrm{se}^2_{\mathrm{phys}}}. \tag{11}$$

The physics term is declared helpful only if $R > \sigma_R$. Otherwise it is reported as non-identifiable. A result that fails this test is reported as failing to clear noise, never as evidence of equivalence. Matched-condition accuracy is reported separately as the difference in matched means. For RQ3 we compare $|\Delta_{\varepsilon_f}|$ with $|\Delta_{\mathrm{RMSE}}|$. The two quantities have different units, so this comparison is indicative only.

Zero-prediction reference. A model outputting $\hat{q} \equiv 0$ has an RMSE equal to the root-mean-square of the reference window. On the three high-variance folds only (On_set1, Off_set1, Off_set2; not computed for the other three folds) a zero prediction scores a window-level RMSE of 1.71–2.05 (group mean over matched test trials).

E. Compute Environment and Latency

The ResNet control ran on an NVIDIA T4 and the pre-trained arm on an RTX 4500 Ada. Runs were replicated across both GPUs from SHA-pinned code; per-run hardware is not a variable in any comparison. Latency is measured at batch size 1 on an input of shape $(1, 2, 5, 224, 224)$, with one warm-up pass, 20 timed passes and explicit CUDA synchronisation. Measured latency is 4–10 ms per window for the ResNet (load-dependent) and 42 ms for the TimeSformer on an RTX 4500 Ada at batch size 1, within the 50 ms budget.

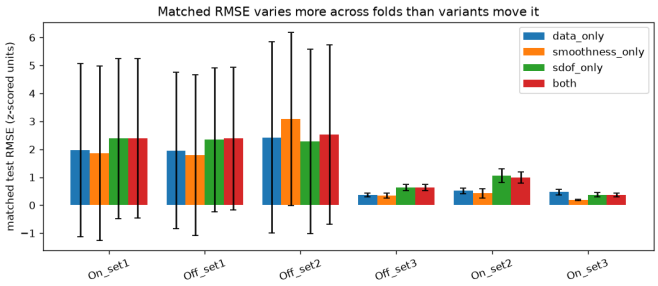


Fig. 1. Matched test RMSE (mean ± trial std) by fold and variant. Variant (b) is smoothness-only.

V. Results and Discussion

A. Matched accuracy (RQ1)

Table I gives the ResNet control results on all six folds for all four variants. Fold labels name the held-out (shift) condition (On/Off = illumination, set1–3 = acquisition set). Each cell is matched RMSE ± trial std over the five matched test trials, with Δ_{RMSE} in parentheses from the three shift test trials. Units are z-scored displacement.

On the three high-variance folds (On_set1, Off_set1, Off_set2), data-only matched RMSE is 1.95–2.43 – at or worse than predicting nothing on the same windows (window-level zero RMSE 1.71–2.05, computed on the three high-variance folds only). Trial stds of 2.8–3.4 on five trials imply at least one test trial with a very large error. Per-trial matched RMSE confirms it: one trial per fold scores 6.9–8.5 (On_set1 7.5, Off_set1 6.9, Off_set2 8.5) with all others at or below 1.6, identically for data-only and both – the same hard trial inflates every variant. On the other three folds, matched RMSE is 0.37–0.52, but Pearson r is only 0.09–0.17. The gain over zero is therefore in amplitude, not waveform shape. The $r > 0.9$ target was missed by a factor of about 3.6.

Matched RMSE varies more across folds than across training variants (Fig. 1). Each fold's matched test set is five of 25 pooled trials, and the split is identical across variants (fixed seed). Fold-to-fold differences therefore plausibly reflect which trials land in test.

Ridge regression on pooled voxel means reaches matched RMSE 0.29–0.66 and r = 0.17–0.24, with $|\Delta| \leq 0.37$. On the three high-variance folds it is far below every deep variant (0.50–0.66 against 1.8–3.1). On the other three folds the comparison is a near tie. Ridge (0.39, 0.40, 0.29) sits beside data-only (0.37, 0.52, 0.47) and smoothness-only (0.34, 0.43, 0.19) point estimates, with deep data-only marginally lower on Off_set3. With $r \approx 0.2$, ridge captures the amplitude envelope and not the shape. In answer to RQ1, no model recovered the LDV waveform under matched conditions; the frequency

TABLE I

MATCHED RMSE (Δ_{RMSE}), 3D-RESNET CONTROL. (A) DATA ONLY, (B) SMOOTHNESS ONLY, (C) SDOF ONLY, (D) BOTH.

Fold	(a)	(b)	(c)	(d)
On_set1	1.98±3.10 (−1.65)	1.86±3.12 (−1.41)	2.38±2.86 (−1.61)	2.39±2.85 (−1.61)
Off_set1	1.95±2.80 (−1.49)	1.79±2.88 (−1.41)	2.34±2.57 (−1.33)	2.39±2.55 (−1.33)
Off_set2	2.43±3.43 (−1.79)	3.08±3.10 (−2.14)	2.29±3.29 (−1.59)	2.53±3.21 (−1.58)
Off_set3	0.37±0.07 (−0.07)	0.34±0.10 (−0.06)	0.64±0.11 (−0.01)	0.63±0.10 (−0.02)
On_set2	0.52±0.10 (+2.23)	0.43±0.17 (+2.23)	1.05±0.25 (+1.94)	0.99±0.21 (+1.98)
On_set3	0.47±0.10 (−0.23)	0.19±0.03 (<0.01)	0.38±0.09 (+0.23)	0.37±0.07 (+0.20)

TABLE II

MAE, MATCHED TEST TRIALS, 3D-RESNET CONTROL. SAME SPLIT AND SEED AS TABLE I; MSE IS THE SQUARE OF THE PER-WINDOW RMSE (EQ. 4) AND RANKS IDENTICALLY, SO IT IS NOT TABULATED SEPARATELY.

Fold	(a)	(b)	(c)	(d)
On_set1	1.80	1.72	2.15	2.16
Off_set1	1.76	1.64	2.09	2.13
Off_set2	2.13	2.66	2.09	2.29
Off_set3	0.29	0.27	0.51	0.51
On_set2	0.41	0.34	0.86	0.80
On_set3	0.38	0.15	0.30	0.30

TABLE III

RIDGECV BASELINE. MATCHED IS MEAN ± TRIAL STD; PEARSON r IS MATCHED.

Fold	Matched RMSE	Shift RMSE	Δ_{RMSE}	r (matched)	MAE
On_set1	0.657±0.713	0.303	−0.354	0.198±0.127	0.539
Off_set1	0.499±0.512	0.322	−0.178	0.201±0.068	0.417
Off_set2	0.637±0.638	0.274	−0.363	0.171±0.084	0.522
Off_set3	0.389±0.098	0.631	+0.242	0.240±0.126	0.309
On_set2	0.398±0.148	0.763	+0.365	0.183±0.036	0.320
On_set3	0.292±0.067	0.358	+0.066	0.232±0.044	0.233

half of RQ1 is addressed in the frequency subsection below. A linear map on pooled voxel features is more robust to a single outlier trial than every deep variant we trained, though on the three low-variance folds the deep variants were equal or better. Trial-level mean Pearson r for (b) is −0.00, 0.17, 0.14, 0.24, 0.14 and 0.25 across folds (On_set1, Off_set1, Off_set2, Off_set3, On_set2, On_set3) – no shape recovery. MSE (RMSE^2) and MAE rank the models identically to RMSE in every fold (ridge MAE 0.23–0.54, deep data-only 0.29–2.13; Table II and Table III), so no conclusion here depends on the choice of error norm.

B. Physics residual and the robustness gap (RQ2)

For data-only, the sign of Δ_{RMSE} is negative on five of six folds (Table I): the held-out condition scored better than the matched pool, which we read as a property of the benchmark. The matched reference is a five-trial mean that mixes five conditions with outlier trials, while each shift set is a single condition. Ridge shows mixed signs (three negative, three positive), and On_set2 is positive for both ridge (+0.37) and deep (+2.23). That fold may be a genuinely harder condition, but its σ_R of about 3 leaves it unresolved.

Under the pre-specified falsification rule (O5), physics failed to clear noise on all six control folds (Fig. 2). Table IV gives the same test for (c) against (a). For (d) against (a), three folds give Off_set1 $R = -0.160$ (σ_R 1.70), On_set2 +0.247 (3.18) and On_set3 −0.434 (0.056).

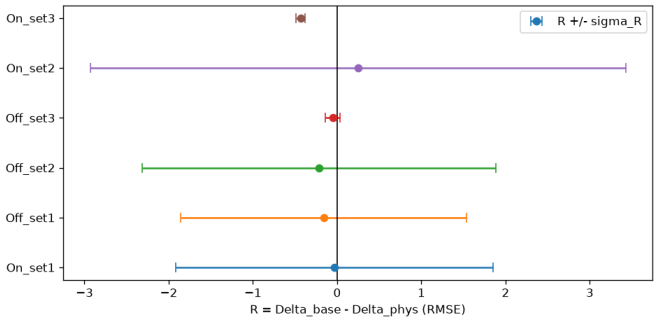


Fig. 2. $R = \Delta(a) - \Delta(d)$ (RMSE) with $\pm\sigma_R$, one point per fold. Physics counts as helpful only if the point lies beyond its bar on the positive side.

TABLE IV

FALSIFICATION TEST (O5): SDOF ONLY (C) AGAINST DATA ONLY (A), RMSE.

Fold	$R = \Delta(a) - \Delta(c)$	σ_R	Clears noise?
On_set1	−0.037	1.886	No
Off_set1	−0.159	1.704	No
Off_set2	−0.200	2.128	No
Off_set3	−0.062	0.087	No
On_set2	+0.295	3.163	No
On_set3	−0.463	0.059	No (harmful side)

On four folds σ_R is 1.7–3.2 RMSE units, so those tests could not detect any plausible effect. Only On_set3 resolves

an effect: its interval excludes zero on the harmful side. Off_set3 has comparable precision ($\sigma_R = 0.087$) but a point estimate of -0.06, smaller than its own noise. For On_set3 the interval excludes zero on the harmful side for both (c) and (d). Physics raised the shifted RMSE from about 0.24 for (a) to about 0.60 for (c) and 0.57 for (d), while slightly lowering matched RMSE. We therefore report failure to clear noise, not equivalence. Note that the rule tests only for benefit, so the On_set3 result is a descriptive observation about direction.

Physics also carries a matched-condition price: relative to (a), (d) raised the matched RMSE point estimate on five of six folds, by 0.11–0.48, and (c) did so on four of six folds, by 0.27–0.54. For On_set3, both lowered it by about 0.1, and (c) also lowered it on Off_set2. On the three high-variance folds the changes lie far inside trial spread (std 2.6–3.4). On Off_set3 and On_set2 the increases of 0.26–0.54 exceed the trial std (0.07–0.25). No noise test was run on this quantity, so we call it a consistent but untested price, not a demonstrated harm. The residual added no measurable per-step cost on a T4 (113.9 against 122.0 ms per window for (c) against (a), within timing noise) and saved no consistent compute.

The smoothness-only variant (b) was trained on every fold. Its matched RMSE point estimate is below (a) on five of six folds (by 0.03–0.28), with the exception of Off_set2 ($+0.65$), and it has the lowest matched RMSE in Table I on On_set3 (0.19 ± 0.03). Its Δ values differ from (a) by at most about 0.35, inside the standard errors on the four high-variance folds. No falsification test was pre-specified for (b), and we make no claim about it beyond these point estimates. Since the smoothness penalty on $\|\hat{q}'\|$ is a shrinkage toward constant output, a lower RMSE under collapse is expected and should not be read as better reconstruction.

C. Kinetics-pretrained TimeSformer, fixed code

Table V gives fixed-code pretrained results (mean-pool readout). Validation Δ uses two shift trials; the On_set1 rows carry matched test RMSE. (ep = epochs-to-target/epochs-run.)

The On_set1 matched-test run logged matched RMSE 1.91 ± 3.50 for (a) and 2.63 ± 3.24 for (d), with Pearson r of 0.08 ± 0.06 and -0.01 ± 0.02 on local re-evaluation – no waveform shape recovery. Both lie within the spread of the control's 1.98 ± 3.10, so pretraining gave no accuracy gain.

D. Frequency and RQ3

Variant (d) with the pretrained encoder reached a relative frequency error of 0.02 on On_set1 while its RMSE was 2.63. This should not be read as frequency estimation. The fit gives $f_d \approx 6.7$ Hz in every fold, so f_n varies little across trials, and a head that outputs a constant would score a low error. A constant baseline predicting the median training-trial f_n (6.73 Hz) scores ε_f 0.001–0.013 on all test sets with $|\Delta| \leq 0.009$, below the trained head's 0.02. Frequency "estimation" here is within reach of a constant, so RQ3 (does frequency error degrade less than RMSE under shift?) remains untested. $|\Delta_{\varepsilon_f}| < |\Delta_{\mathrm{RMSE}}|$ held on all six control folds, but most Δs are negative and the two quantities have different units. We regard RQ3 as untested.

E. Failure analysis

On every fixed-code run the data-only model collapses to near-constant output by epoch 1–6 (pretrained 1–11), and matched RMSE (1.9–2.6 on the high-variance folds) sits at or above the zero-prediction reference (1.71–2.05). The faster convergence of the pretrained arm is best read as reaching the collapsed solution sooner, not as successful transfer.

A second failure is frequency without amplitude: physics-trained models match the oscillation prior, but amplitude is anchored only by L_{LDV}, since L_{phys} is amplitude-blind by design.

The latents are impoverished: probes on the (d) latents (On_set1, 66 test windows) gave $R^2 \approx -0.2$ for f_n and 0.51–0.57 for envelope slope (vs ≈ 0 shuffled). Cross-condition latent cosine was about 1.0 and CKA 0.03–0.05. The encoder carries envelope energy, not oscillation. The 66 windows come from few trials and are correlated, so these R^2 values are indicative only.

Self-consistency score as an uncertainty proxy. We proposed the normalised residual as a confidence score. We correlated it with per-window error on 82–90 windows per fold. For the pretrained models the correlation was weak or undefined ($|r| < 0.3$, e.g. 0.19 and 0.09 for Off_set2 (d)). For ridge the Spearman correlation was negative on all folds checked (-0.62 to -0.84), so larger residuals went with smaller errors, the wrong direction for an uncertainty score. Since L_{phys} is amplitude-blind and dominated by noise, this is unsurprising, but it means the residual is not usable as calibrated confidence here.

Why physics did not help (hypothesis). L_{phys} sits at 10^{11}–10^{12} and is 3–41$\times$ smaller for physics-trained models on five of six folds, but larger on On_set3. It is dominated by $1/dt^2$ amplification of high-frequency noise in $\hat{q}$, while the targets are interpolated from 256 Hz and carry no energy above 128 Hz. The residual may therefore act as a high-frequency suppressor rather than a dynamics prior. This is untested. It differs from the gains in [7], where mass, damping and stiffness were known. Here k and c come from one trial, and a poorly specified residual can harm accuracy [6].

F. Limitations and significance

The study uses one rig, 30 trials, five matched and three shift test trials per fold, and a 6.7 Hz mode sampled at 256 Hz, so the sensor's kilohertz capability is untested. The 5-bin voxel integrates events over 200 ms per bin, a 5 Hz sampling rate with a 2.5 Hz Nyquist limit, so the 6.7 Hz mode aliases to 1.7 Hz and the input cannot represent the waveform's phase; the $B \in \{3, 5, 7, 9\}$ ablation (at most 9 Hz) does not address this limit. Waveform-shape and physics results are therefore upper bounds set by this representation, not tests of event-camera capability at adequate temporal resolution. σ_R is too large on four folds to detect any plausible effect. Seed variance was checked only for the ResNet in a two-fold study.

TABLE V
PRETRAINED TIMESFORMER, FIXED CODE (MEAN-POOL READOUT).

Fold / run	data valΔ (ep)	both valΔ (ep)	data matched	both matched
Off_set3	-3.05 (2/15)	-2.68 (1/11)	—	—
On_set1	$+0.49$ (1/11)	$+0.33$ (5/25)	—	—
On_set2	$+0.18$ (11/21)	$+0.14$ (3/20)	—	—
On_set1, voxel fp16	$+0.50$ (1/11)	$+0.34$ (6/22)	1.91 ± 3.50	2.63 ± 3.24
On_set1, time surface	$+0.29$ (1/13)	$+0.33$ (6/23)	2.09	2.61
On_set1, B = 3/5/7/9 (both)	—	B=3/5/7/9: 0.327/0.332/0.328/0.330	—	—

Pretrained coverage, probing and calibration rest on single folds or checkpoints. The result that survives is narrow. Under this protocol, an SDOF residual did not improve robustness, a ridge baseline is a competitive bar, and the benchmark may need more trials before Δ can discriminate methods.

VI. ABLATION ANALYSIS

Beyond the four loss variants of Table I and the O5 test of Fig. 2, we ablated four implementation factors on the single On_set1 fold: the number of temporal bins B, the input representation (voxel against time surface), numerical precision and seed variance.

Validation Δ of (d) on On_set1 is 0.327/0.332/0.328/0.330 for B = 3/5/7/9. For (d) the time-surface run (0.33) and the fp16 run (0.34) match the fp32 run (0.33). For (a) the values are 0.49 (fp32), 0.50 (fp16) and 0.29 (time surface). The surface difference for (a) is about 0.2, we do not interpret it. These comparisons are single-fold and single-seed.

B, input representation and precision moved the (d) validation Δ by at most about 0.01. A bounded two-fold, two-seed ResNet check showed seed-to-seed differences of about 1.0 (Off_set1, data only: -2.77 vs -1.80) to 2.85 (Off_set2, data only: -3.15 vs -0.30; both: -2.62 vs -0.17). Seed variance of the TimeSformer arm was not measured, so the comparison across backbones is indicative. Even so, differences below 0.01 are far smaller than the seed noise we did observe. These uninformative nulls show insensitivity within a collapsed regime, not that B, representation or precision are generally irrelevant.

VII. CONCLUSION AND FUTURE WORK

We tested whether a single-degree-of-freedom (SDOF) physics residual makes event-based vibration estimation more robust to changes in sensing condition. The test used 30 hammer-impact free-decay trials of one H-beam, six lighting/set conditions, trial-level leave-one-condition-out evaluation, and a pre-specified rule that the residual must reduce the robustness gap Δ by more than its standard error.

The residual failed that rule on all six control folds. On four folds the standard error was too large to detect any plausible effect. Only On_set3 resolved an effect, and there the interval excluded zero on the harmful side. The residual raised the matched RMSE point estimate on most folds, though that quantity was not tested for significance.

No model recovered waveform shape. Pearson r stayed at or below 0.25 against a target above 0.9. Deep models often collapsed to near-constant output within a few epochs. A ridge regression on pooled voxel features was more robust to a single outlier trial than every deep variant we trained, but not more accurate on the three low-variance folds (0.29–0.40 vs 0.19–0.52), and it captures amplitude envelope only. Latent probing found envelope decay but not natural frequency. Several Δ values were negative, which we read as an artefact of the matched reference being a five-condition mixture at five test trials, so Δ confounds composition with shift, not as evidence that shifted conditions are easier.

The main contribution is methodological. Δ with group-mean standard errors, trial-level splits and a falsification rule make a regulariser's benefit checkable for continuous-waveform targets, where classification accuracy does not apply. Under this protocol, a physics prior on a small dataset should not be assumed to help, and a simple linear baseline should be reported with any deep event-to-waveform model. The negative result also suggests that the benchmark, not just the model, may limit what can be concluded.

The study covers one rig, a 6.7 Hz mode sampled at 256 Hz and few trials per condition. The residual was identified from a single trial per fold, and the explanation we offer for the physics null, that it acts as a high-frequency suppressor, is a hypothesis we did not test.

Future work should:

- Collect more trials and conditions, over a wider frequency range, so that σ_R is small enough to detect effects of practical size.
- Evaluate on faster, higher-frequency structures where event cameras have a real advantage over frame cameras and the 128 Hz bandwidth limit of the 256 Hz LDV reference no longer applies.
- Use amplitude-anchored or two-stage training, with frequency fitted first and scale second, since the physics term is amplitude-blind.
- Add frequency-aware inputs or pretraining, because probing showed the latents lack natural frequency.
- Test the high-frequency-suppression hypothesis directly, for example by band-limiting the residual or by estimating k and c jointly with the network.
- Report trivial and linear baselines, including a constant-frequency head and window-level zero prediction, as standard practice in this task.

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Data and Code Availability

- *Source Code:* The implementation developed for this study is publicly available on GitHub.
- *Primary Dataset:* The dataset supporting the findings of this work is archived in Zenodo (Record 17447083, version v1; https://doi.org/10.5281/zenodo.17447083) under the Creative Commons Attribution 4.0 International (CC BY 4.0) license.
- *Kaggle Mirrors and Kernels:* The public dataset and reproducible training kernels are hosted on Kaggle (dataset, kernels).

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